

Brain cancer associated with environmental lead exposure: evidence from implementation of a National Petrol-Lead Phase-Out Program (PLPOP) in Taiwa...

BACKGROUND AND OBJECTIVE: In 1981, a Petrol-Lead Phase-Out Program (PLPOP) was launched in Taiwan for the abatement of environmental lead emissions. The present study was intended to examine whether the high Petrol-Lead Emission Areas (PLEA) would result in an increase in the incidence rate of brain cancer based on a national data bank. **METHODS:** The national brain cancer incidence data was obtained from the Taiwan National Cancer Registry. Age standardized incidence rates were calculated based on the 2000 WHO world standard population, and gasoline consumption data was obtained from the Bureau of Energy. The differences in the trend tests for age-standardized incidence rates of brain cancer between high, median, low, and small PLEA were analyzed. **RESULTS:** A significant increase was found from small to high PLEA in age-standardized incidence rates of brain cancer. By taking six possible confounders into account, the age-standardized incidence rates for brain cancer were highly correlated with the median and high PLEA by reference to the small PLEA. **CONCLUSION:** After being adjusted for a number of relevant confounders, it could be concluded that high PLEA might result in an increase in the incidence rate of brain cancer resulting from high lead exposures. Copyright © 2011 Elsevier Ltd. All rights reserved.